

Design & Analysis of Algorithms

Lab 1 Assessment

Name: Akanksha Kumari

Roll No: 2501010076

Email: 2501010076@krmu.edu.in

Q1. 1.7 Complete Algorithm Performance Assessment (10 marks)

```
def complete_algorithm_performance_assessment(n, arr, target):
    def factorial_recursive(x):
        if x<=1:
            return 1
        return x*factorial_recursive(x-1)
    def factorial_iterative(x):
        res=1
        for i in range(2,x+1):
            res*=i
        return res
    def fibonacci_recursive(x):
        if x<=0:
            return 0
        if x==1:
            return 1
        return fibonacci_recursive(x-1)+fibonacci_recursive(x-2)
    def fibonacci_iterative(x):
        if x<=0:
            return 0
        if x==1:
            return 1
        a,b=0,1
        for _ in range(2,x+1):
            a,b=b,a+b
        return b
    def bubble(arr):
        for i in range(n):
            for j in range(1,n-i-1):
                if arr[j+1]>arr[j]:
                    arr[j+1],arr[j]=arr[j],arr[j+1]
    def insertion(arr):
        arr=list(arr)
        n=len(arr)
        comp=0
        shift=0
        for i in range(1,n):
            key=arr[i]
            j=i-1
            while j>=0:
                comp+=1
                if arr[j]>key:
                    arr[j+1]=arr[j]
                    shift+=1
                    j-=1
                else:
                    break
            arr[j+1]=key
        return arr,comp,shift
    def linear(arr,target):
        comp=0
        idx=-1
        for i in range(len(arr)):
            comp+=1
            if arr[i]==target:
                idx=i
                break
        return comp,idx
    def run_ass(n,arr,target):
        print("Algorithm Performance Assessment")
        print("Computation Results")
        print(f"Factorial Recursive: {factorial_recursive(n)}")
        print(f"Factorial Iterative: {factorial_iterative(n)}")
        print(f"Fibonacci Recursive: {fibonacci_recursive(n)}")
        print(f"Fibonacci Iterative: {fibonacci_iterative(n)}")
        b_sort,b_comp,b_swap= bubble(arr)
        l_idx,l_comp= insertion(arr)
```

```
print("Search Results")
print(f"Linear Index: {l_idx}")
print(f"Linear Comparisons: {l_comp}")
print(f"Binary Index: {b_idx}")
print(f"Binary Comparisons: {b_comp}")
print("Search Best: Binary Search")
print("Sorting Results")
print(f"Bubble Sorted: {bubble(arr)}")
```